

ASSIGNMENT 5

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Q1: Explain the difference between Manual Testing and Automation Testing. What are the main advantages of automation testing?

Difference between Manual Testing and Automation Testing:

Manual Testing	Automation Testing
Test cases are executed manually by a tester	Test cases are excuted using automation tools/scripts
More time-consuming and prone to human errors	Faster, more accurate, and reusable
No coding needed	Coding knowledge required
Good for exploratory and usability testing	Best for repetitive and regression testing
Testing tools: JIRA, TestRail, Zephyr	Testing tools: Selenium, Cypress, Appium, Playwright, TestNG/Junit, Jmeter, Cucumber

Main advantages of Automation Testing:

- Faster execution of test cases.
- Improves accuracy by reducing human errors.
- Supports repeated regression testing.
- Saves time and effort in the long run.

Q2: Describe the steps involved in setting up a Selenium test environment (installation, configuration, and browser setup).

1. Install **Java JDK** and set the `JAVA_HOME` environment variable.
2. Install an IDE such as **Eclipse** or **IntelliJ IDEA**.
3. Download and add **Selenium WebDriver** libraries to the project.
4. Download the required browser driver.
5. Configure the driver path in the Selenium script.
6. Install and launch the browser to execute Selenium tests.

Q3: Write a simple Selenium WebDriver script in Java to open Google, search for “Automation Testing,” and close the browser.

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class GoogleSearch {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();

        driver.get("https://www.google.com");

        WebElement searchBox = driver.findElement(By.name("q"));
        searchBox.sendKeys("Automation Testing");
        searchBox.submit();

        driver.quit();
    }
}
```

Q4: What are the main components of Selenium? Explain the purpose of Selenium Grid with an example use case.

1. Selenium IDE
2. Selenium WebDriver
3. Selenium Grid

Purpose of Selenium Grid:

- Selenium Grid allows tests to run on multiple browsers, operating systems, and machines simultaneously.
- It supports parallel execution, reducing test execution time.

Example Use Case:

A company wants to test its website on **Chrome, Firefox, and Edge** at the same time. Selenium Grid executes the same test script across all browsers simultaneously, saving time.

Q5: Using Selenium IDE, record a login test case for any demo website. List the key steps and explain how “Record and Play” works in Selenium IDE.

Demo Website: <https://practicetestautomation.com/practice-test-login/>

Login Test Steps:

1. Open Selenium IDE.
2. Click Record a New Test.
3. Enter the website URL.
4. Enter username: student.
5. Enter password: Password123.
6. Click the Login button.
7. Verify successful login message.
8. Stop recording and save the test case.

How "Record and Play" Works in Selenium IDE:

- **Record:** Selenium IDE captures user actions such as clicks, typing, and navigation as test commands.
- **Play:** Selenium IDE automatically re-executes the recorded commands and verifies whether the application behaves as expected.